

Release notes

Repository source: RELEASE_NOTES.md

Release candidate v1.0.0-rc1 (unreleased)

This is the release-candidate record for a proposed computer-assisted universal lower bound in the three-dimensional Blaschke–Lebesgue problem. The candidate metadata and licensing are fixed. The audited candidate has passed a three-job hosted workflow and a complete named-author fresh-clone replay, but no public release or immutable tag is asserted yet.

Proposed result

For every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$, the package represents the proposed inequality

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.4118775637803022742481789259828398710 \dots d^3.$$

The result is universal: no symmetry, smoothness, contact-type, or fixed circumradius assumption appears in the final theorem.

Included in the draft package

- a ten-page LaTeX manuscript and reference PDF;
- one standalone exact-rational certificate;
- immutable JSON, NPZ, TSV, and CSV proof objects;
- an independent NumPy binary64 audit;
- an independent base-R audit;
- ten reconstruction and regression tests;
- seven adversarial proof-object mutation tests;
- automatic paper/certificate and appendix/certificate reconciliation;
- reproducibility, trust-boundary, claims, source, audit, and proof ledgers;
- paired Markdown/PDF documentation;
- SHA-256 manifests for the repository, certificate, paper, and rendered docs;
- a three-job GitHub Actions workflow for Python, R, and PDF/document checks;
- final candidate authorship, repository, citation, and split-license metadata.

Proof architecture

The proof splits the normalized circumradius interval at

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

The lower branch is a gap-free exact radial certificate whose unique bottleneck is 03B02. In the near-Jung branch, four circumsphere contacts force a 98,306-ray core. An exact determinant transfer, a dominant-edge coordinate-width estimate, and three optimized antipodal cap pairs give

<pre>near-Jung lower = 0.4118796712148324317327979810531508741628745212952960018 handoff margin = 0.0000021074345301574846190550703110031602996750911392740</pre>
--

so the lower-row coefficient is the universal bottleneck.

Verification status

Completed in recorded environments:

- exact finite verification and byte-identical certificate reconstruction;
- independent Python binary64 audit;
- independent base-R audit on the named author’s machine and hosted R 4.6.1;
- ten regression tests and seven mutation tests;
- paper/certificate and exact-appendix reconciliation;
- manuscript and 22 Markdown/PDF builds;
- hosted Poppler preflight of committed and rebuilt PDFs;
- a private staging-remote push of candidate commit `892d4bc6fad07950e78da66e45b95063a6415af6`;
- three green hosted jobs at <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>;
- a complete named-author fresh-clone replay under Python 3.14.7, NumPy 2.3.5, and R 4.6.1; and
- a clean fresh-clone working tree and complete five-commit bundle.

Still required before release:

- green hosted jobs on this evidence-bearing commit;
- final named-author copy and page approval;
- final release date and transition from `v1.0.0-rc1` to `v1.0.0`;
- immutable release assets and hashes;
- green hosted jobs and a clean replay from the exact final tag; and
- explicit named-author approval of the final paper, README, release notes, tag, visibility change, GitHub Release, and announcement.

Candidate publication metadata

- sole named author: Bruce J. Swihart;
- no institutional affiliation asserted;
- private staging repository and intended canonical repository: <https://github.com/swihart/minvol-degree3-contact-bound>;
- audited candidate commit: `892d4bc6fad07950e78da66e45b95063a6415af6`;
- preferred citation: `CITATION.cff`;
- paper and prose license: CC BY 4.0;
- software and build-infrastructure license: MIT;
- candidate version: `v1.0.0-rc1`;
- target final tag: `v1.0.0`;
- final release date and archival identifier: not yet assigned.

Scope and limitations

This draft does not prove:

- sharpness or equality;
- rigidity;
- the identity of a minimizing body;
- the Meissner conjecture;
- independent reproduction or peer review.

The Meissner value is an explicit-body comparison value, not a universal minimum established here.

Relationship to prior work

The package builds on classical constant-width geometry, the support-function and spectral perspective developed in recent work, and the direct contact-geometric exact-rational architecture of the public HYRA predecessor. The repository has fresh history and does not alter the earlier public MinVol spectral-bound repository.

A dated source and recent-literature check is recorded in [proof/SOURCE_LEDGER.md](#). It is targeted rather than exhaustive, so this draft avoids an unqualified “best known” claim.

Corrections and versioning

A change to any theorem-bearing proof object, exact verification rule, radius partition, or coefficient requires a new version and explicit mathematical comparison. Corrections must be recorded in release notes; released history must not be silently rewritten.